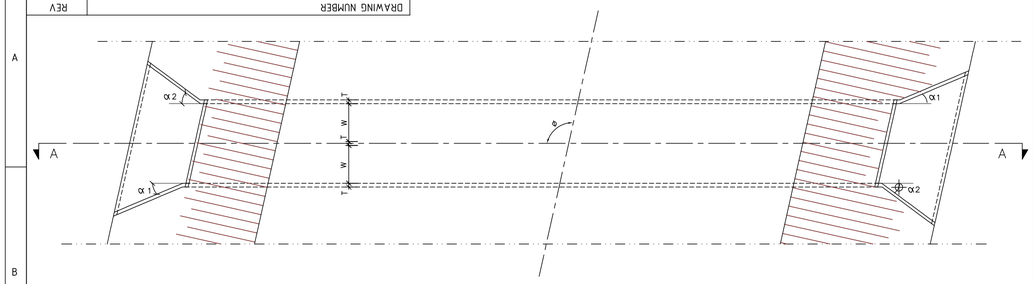
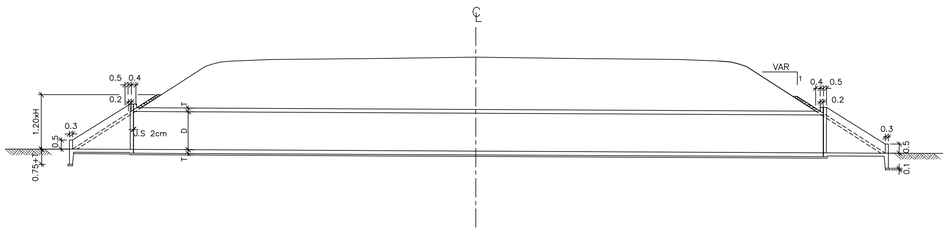
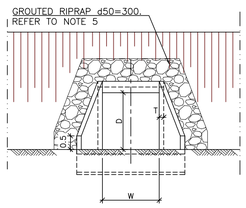
DRAWING NUMBER MD - 216 - 49EA - EG - CV - DED - 1003 REV D01



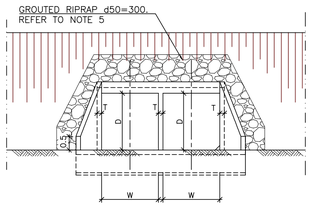
PLAN



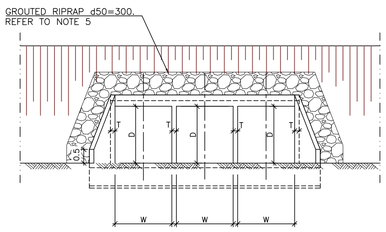
SECTION A-A



ELEVATION
SINGLE CELL BOX CULVERT

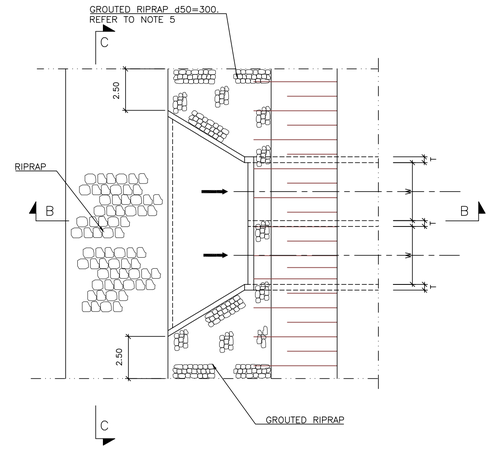


ELEVATION
DOUBLE CELL BOX CULVERT

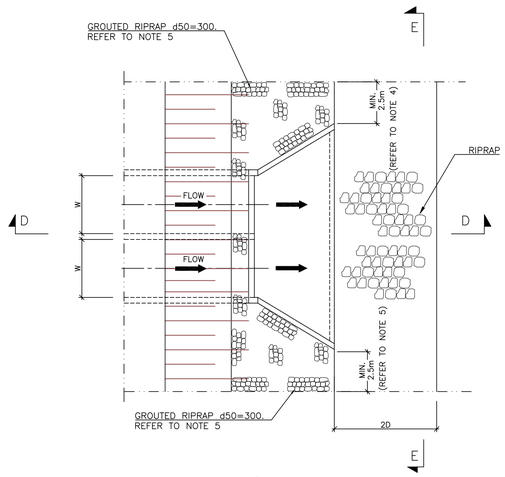


ELEVATION
TRIPLE CELL BOX CULVERT

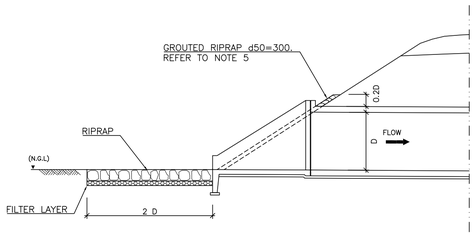
CULVERT TYPICAL DETAILS



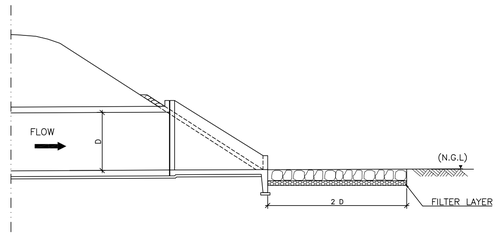
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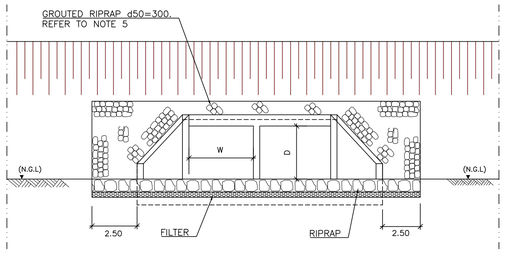
PLAN
CULVERT OUTLET



SECTION B-B

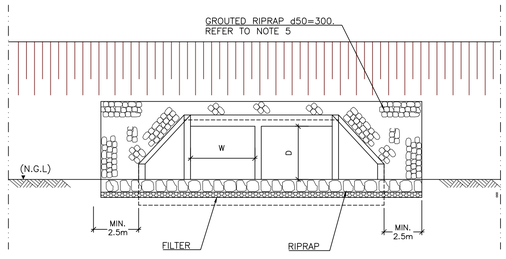


SECTION D-D



SECTION C-C

CULVERT INLET AT NATURAL GOUND LEVEL



SECTION E-E

CULVERT OUTLET AT NATURAL GOUND LEVEL

NOTES:-

1- ALL DIMENSIONS ARE IN METERS UNLESS OTHERWISE SPECIFIED.

2- ALL DRAWINGS SHOULD FOLLOW THE MOT STANDARD DRAWINGS.

3- FOR STRUCTURAL DETAILS REFER TO STRUCTURAL DETAILS DRAWING:

4-REFER TO THE PLAN-PROFILE DRAWINGS FOR THE EXTENT OF THE PROTECTION. IF THIS EXTENT IS NOT DEFINED ON THE DRAWINGS THEN 2.5m EXTENT FROM BOTH SIDES OF THE CULVERT SHOULD BE PROVIDED.

5-ALL CULVERT'S INLETS AND OUTLETS SHOULD HAVE GROUTED RIPRAP OF SIZE d50=300 mm EXCEPT THE OULET OF CW09 d50=400 mm.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	08/09/2023	AS-BUILT	--	SNM	--	SJM	
		C01	02/06/2020	ISSUED FOR CONSTRUCTION AND REVISED AS PER EMPLOYER COMMENT ON 13/05/2020	--	SNM	--	SJM	
		A01	16/04/2020	ISSUED FOR REFERENCE	--	SNM	--	SJM	

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Saudi Arabian Mining Company شركة التعدين العربية السعودية
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DRAWING NUMBER	SA1930-PK1-DD-WS-501		REVISION 0
DATE	10/04/2020	DATE 11/04/2020	DATE 12/04/2020
DRAWN	Y.A	CHECKED M.S	APPROVED M.E

LARSEN & TOUBRO DRAWING NUMBER	019045_910_C_V_24019_003	REVISION 2
OUTOTEC DRAWING NUMBER	900707__GAU01__AG90__CTB17__00003	DOCUMENT ID OU100681671 REVISION 2

DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SNM	13/04/2020	DETAIL DRAWING
ENGINEER	--	--	STORMWATER TYPICAL DETAILS-ACCESS ROAD TO MANSOURAH-MASSARAH PROCESS PLANT
APPROVED	SJM	16/04/2020	SHEET 1 OF 3
EMPLOYER			
SCALE	N.T.S	A1	DRAWING NUMBER MD - 216 - 49EA - EG - CV - DED - 1003 REV D01

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

PROJECT: MANSOURAH MASSARAH GOLD PROJECT — **PROJECT NO.: 216**
STAGE: 4 - PROJECT EXECUTION. — **REVIEW LOG DOC. NO. & REV.:**
ACONEX REF. NO. & DATE: BECHTEL-WTRAN-002458, 13 May 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 13 May 2020	Comment by	EPC Response DATE: 02 June 2020	Status	Remarks
			General Comments (cc)				
MD-216-49EA-EG-CV-DED-1003-A01	49EA	1	Please provide antecedents for the Grouted Riprap (size of stones, morter, etc.)	Mr. Doron Golan BECHTEL	Incorporated in the drawing. Please see MOC General specification - November 1998, section - 6.06 attached below.	Closed	

SECTION 6.06 - RIPRAP AND SLOPE PROTECTION

6.06.1 DESCRIPTION. **This Work shall consist of the placement of riprap, concrete, gabions, wire mesh, or other forms of protection on slopes at the locations shown on the plans, or as ordered by the Engineer, in accordance with the specifications and in conformity with the lines, grades, thickness and typical cross sections shown on the plans or established by the Engineer.**

ITEMS IN BILL OF QUANTITIES
Loose Stone Riprap
Grouted Stone, Riprap
Sacked Concrete Revetment
Gabions
Reinforced Concrete Slope Protection
Wire Mesh Slope Protection
Paving Tile Slope Protection
Wire Enclosed Riprap Slope Protection

6.06.2 MATERIALS.

6.06.2.1 Stone for Riprap. **Stone for grouted and loose riprap shall be hard, sound and durable, furnished in broad flat shapes to the maximum extent practicable. Samples of the stone to be used shall be submitted to and approved by the Engineer before any stone is placed.**

The minimum apparent specific gravity shall be two and five tenths (2.5) and the maximum absorption shall be six (6) percent when tested in accordance with MRDTM 304. The stone shall not have an abrasion loss greater than forty-five (45) percent when tested in accordance with MRDTM 309.

The weight of stone for the various classes of riprap shall be as follows:

TABLE 6.06-1

Size of Stone (Kilograms)				Percent of the Total Weight Smaller than Size Shown
Class A	Class B	Class C	Class D	
50	200	1,000	5,000	95-100
20	100	500	2,000	50-100
5	20	100	500	0-50
1	5	20	100	0-10

Stone for "Loose Stone Riprap" shall be quarried, fractured, or otherwise predominantly angular. Stone for other riprap and gabions may be rounded (boulders) or angular.

6.06.2.2 Grout. Grout for "Grouted Stone Riprap" shall conform to Subparagraph 5.01.3.1.2, "Mortar" in these General Specifications.

6.06.2.3 Concrete. Concrete for "Sacked Concrete Revetment" shall conform to Subsection 5.03.9, "Concrete for Minor Structures" in these General Specifications except that aggregates may be produced in and batched from a single stockpile and the minimum cement content shall be five and one-half (5½) sacks per cubic meter.

Concrete for "Reinforced Concrete Slope Protection" shall conform to Class A in Section 5.01, "Portland Cement Concrete" in these General Specifications.

6.06.2.4 Sacks. Sacks for "Sacked Concrete Revetment" shall be of minimum two hundred (200) grams per square meter burlap. They shall be approximately six hundred (600) millimeters by nine hundred (900) millimeters measured when empty. Relatively clean reclaimed sacks may be used if approved by the Engineer.

6.06.2.5 Gabions. Gabion baskets shall be constructed of galvanized steel wire mesh conforming to ASTM A 390, Class 3, and the requirement shown on the plans. The wire mesh shall be twisted to form hexagonal openings of uniform size. The maximum nominal opening size shall be one hundred (100) millimeters. The mesh shall be constructed so as to resist pulling apart at any of the twists or connections forming the mesh when a single wire strand in a section is cut.

Baskets shall be supplied in one or more of the sizes shown on the plans so as to be assembled forming the minimum dimensions, stability, and structural integrity of the type of installation specified.

Baskets shall be fabricated in such a manner that the sides, ends, lid (except for gabion blankets), and diaphragms can be assembled at the construction site into rectangular baskets of the specified sizes. Gabion baskets shall be of single unit construction-base, lid, ends, and sides shall be either woven into a single unit or one edge of these members connected to the base section of the gabion in such a manner that strength and flexibility at the point of connection is at least equal to that of the mesh.

All perimeter edges of the mesh forming the basket shall be securely clip bound or selvedged so that the joints formed by tying the selvedges have at least the same strength as the body of the mesh.

Perimeter (edge), tie, and connection wires shall conform to ASTM A 641, Class 3, Medium Temper, and the minimum diameters shown on the plans.

Alternate equivalent designs of gabion baskets may be submitted to the Engineer for approval.

Riprap fill material for gabions shall be Class A as shown in Table 6.06-1.

6.06.2.6 Reinforcing Steel. Reinforcing steel for "Reinforced Concrete Slope Protection" shall conform to Subsection 5.02, "Reinforcing Steel" in these General Specifications.

6.06.2.7 Wire Mesh Slope Protection. Wire mesh for "Wire Mesh Slope Protection" shall be chain-link fence fabric conforming to ASTM A 491 or ASTM A 392. Minimum wire size shall be four and eight tenths (4.8) millimeter diameter (6 AWG).

Galvanized wire rope shall conform to ASTM A 603 and shall be the size shown on the plans.

Rock bolts, fasteners, and hardware shall be galvanized and conform to the requirements shown on the plans.

6.06.2.8 Wire Enclosed Riprap Slope Protection. Wire fabric shall meet the requirements of ASTM A 116, Class 2, minimum four (4) millimeters diameters (No. 8 AWG). Tie wires shall conform to ASTM A 641, Class 3, soft or medium temper, minimum four (4) millimeters diameter.

Piles, deadmen, and other hardware shall conform to the requirements shown on the plans.

Riprap for "Wire Enclosed Riprap" shall be Class A as shown in Table 6.06-1.

6.06.2.9 Geotextile Fabric. Geotextile Fabric shall be a woven or nonwoven geotextile meeting the following requirements of Section 6.16, "Geotextile Construction" in these General Specifications for Type 1 Fabric-Geotextile for Slope Protection and Channel Lining.

6.06.2.10 Joint Filler. Preformed joint filler shall conform to the requirements of AASHTO M 33. Hot-poured joint filler shall conform to AASHTO M 282.

6.06.3 CONSTRUCTION REQUIREMENTS. The bed upon which the riprap or slope protection is to be placed shall be excavated to the required grades and lines as shown on the plans or as directed and approved by the Engineer. A footing trench shall be excavated along the toe of the slope as shown on the plans or as directed by the Engineer. All footing trenches and excavations shall be approved by the Engineer prior to placement of stones or concrete. Subgrade or base should be firm or compacted as approved by the Engineer. Stones shall be placed so as to provide a minimum number of voids; larger stones shall be placed in the footing trench and on the outside surface of the slope.

6.06.3.1 Loose Riprap. The loose riprap shall be placed by dumping and spreading in layers by a mechanical device or other methods approved by the Engineer, all to secure a stable mass. After the completion and approval of the riprap placement, the

surface voids of the riprap in the footing trench and on the lower portions of the slope shall be filled with excavated material and dressed to the satisfaction of the Engineer.

6.06.3.2 Grouted Riprap. Riprap shall be arranged in such a way that the largest rocks are at the bottom of the slope. The surfaces of the rock shall be cleaned of adhering dirt and clay.

The space between stones of grouted riprap shall be filled with concrete mortar. After riprap placement, the grout shall be vibrated, spaded, and rodded into place until the voids are completely filled to a minimum depth of five hundred (500) millimeters from the face of the riprap. Excess material and spillage shall be cleaned from the front face of the riprap before hardening. The exposed mortar shall, immediately after completion of a section of riprap, be cured using clear curing compound in accordance with Subparagraph 5.03.4.10.1, "Membrane Curing" in these General Specifications.

6.06.3.3 Sacked Concrete Revetment. After the slopes have been prepared as shown on the plans, and have been approved by the Engineer, the sacked concrete riprap shall be placed thereon. The sacks shall be filled with concrete loosely placed so as to leave room for tying and sewing. Approximately thirty-five thousandths (0.035) cubic meter of concrete shall be placed in each sack. The sacks shall be closed by sewing tied with strong twine. Immediately after closing, the sacks shall be placed and trampled lightly to cause them to conform to the earth surface and the adjacent sacks already in place. The sacks shall be laid in accordance with the details shown in the plans. All dirt and debris shall be removed from the top of the sacks before the next course is laid. Sacks shall be placed so that the tied ends will not be adjacent to one another. No more than six (6) courses of sacks shall be placed in any tier until initial set in the first course of any such tier has taken place.

In placing sacks that are filled with concrete, care in shaping the sacks shall be exercised, to the end that the minimum dimensions shown on the plans will be obtained, and also that the finished sacked concrete revetment will have a minimum of voids.

6.06.3.4 Gabions. Each gabion unit shall be assembled by binding together all vertical edges with wire ties on approximately one hundred fifty (150) millimeter spacing or by a continuous piece of connecting wire stitched around the vertical edges with a coil about every one hundred (100) millimeters. Empty gabion units shall be set to line and grade as shown on the plans or as directed by the Engineer. Wire ties or connecting wire shall be used to join the units together in the same manner as described above for assembling. Internal tie wires shall be uniformly spaced and securely fastened in each cell of the structure.

Assembly details of gabions shall be as specified in the HDM, Volume 4, Standard Drawings SP-4 through SP-7.

Where the length of the gabion exceeds its horizontal width the gabion shall be equally divided by diaphragms, of the same mesh and gauge as the body of the gabions, into cells whose length does not exceed the horizontal width. The gabion shall be

furnished with the necessary diaphragms secured in proper position on the base section in such a manner that no additional tying at this juncture will be necessary.

The last lift of stone in each cell shall be level with the top of the gabion in order to properly close the lid and provide and even surface for the next course.

All gabion units shall be tied together each to its neighbor along all contacting edges in order to form a continuous connecting structure. Empty gabions stacked on filled gabions shall be laced to the filled gabion at the front, side and back.

A standard fence stretcher, chain fall, or iron rod may be used to stretch the wire baskets and hold alignment.

The gabions shall be filled with stone carefully placed by hand or machine to assure alignment and avoid bulges with a minimum of voids. Alternate placing of rock and connection wires shall be performed until the gabion is filled. After a gabion has been filled, the lid shall be bent over until it meets the sides and edges. The lid shall then be secured to the sides, ends, and diaphragms with the wire ties or connecting wire in the manner described above for assembling.

6.06.3.5 Reinforced Concrete Slope Protection.

6.06.3.5.1 Surface Preparation. Slopes to be protected shall be compacted and finished to grade prior to the installation of reinforcing steel, bulkheads, and joint material. The Contractor shall furnish screed rails or other grade indicating devices to control the minimum depth of concrete and uniformity of the top surface.

6.06.3.5.2 Reinforcing Steel and Joints. Reinforcing steel shall be supported firmly so as not to be subject to displacement during construction. Joints shall be laid out or referenced accurately. Construction joints shall be placed at designated contraction joints, except they shall be expansion type whenever constructed more than five (5) meters from a free edge or another expansion joint.

6.06.3.5.3 Concrete Placement. Concrete shall be placed and finished in accordance with the requirements of Section 5.03, "Concrete Structures" in these General Specifications. The Contractor may select a uniform floated or broomed finish approved by the Engineer. Joint blockouts for poured joint filler may be formed in place or inserted in the fresh concrete.

6.06.3.5.4 Curing. Concrete shall be cured in accordance with the requirements of Paragraph 5.03.4.10, "Curing and Protection" in these General Specifications using membrane or water curing.

6.06.3.6 Wire Mesh Slope Protection. Wire Mesh Slope Protection shall be installed in accordance with the typical details shown on the plans, and at locations shown on the plans or ordered by the Engineer.

In particular, wire mesh slope protection shall be placed as shown in the HDM, Volume 4, Standard Drawing SP-1, depending on the slope, its nature, and to the sizes of loose boulders, if any.

Wire mesh shall be folded onto itself for thirty (30) centimeters minimum, and firmly anchored on the tope of the slope. Anchors may be expansion bolts or reinforcing bars, driven or bonded to the slope with non-shrink grout or encased in concrete. Type, size, length and spacing of anchors and anchor rods shall be as shown on the plans.

Before laying the wire mesh, all loose rock and boulders shall be removed and disposed of as approved. The cable grid shall be laid, the wire mesh sheets shall be rolled down the slope, secured to the cables, and laced together using galvanized steel wire two gauges heavier than that used in the main body of the mesh. Where overlapping occurs, the sheets shall be overlapped without cuts.

Where shown on the plans or directed by the Engineer, wire mesh shall be anchored to the slope, using one anchor every fifteen (15) square meters of mesh. The toe of the slope protection shall be anchored or left loose, as shown or as directed.

6.06.3.7 Paving Tile Slope Protection. Paving Tile Slope Protection shall be installed in accordance with the Plan details and as detailed in Paragraph 6.12.3.2 "Precast Tiled Sidewalks' in these General Specifications.

6.06.3.8 Wire Enclosed Riprap Slope Protection. Wire Enclosed Riprap Slope Protection shall be constructed in accordance with the typical details shown on the plans, and at locations shown on the plans or ordered by the Engineer.

6.06.3.9 Geotextile Fabric. Geotextile Fabric shall be furnished and installed, when required, by the typical details shown on the plans and as detailed in Section 6.16, "Geotextiles" in these General Specifications for Type I Fabric-Geotextile for Slope Protection and Channel Lining.

6.06.4 QUALITY ASSURANCE PROCEDURES. The riprap and slope protection will be inspected, sampled, tested and evaluated in accordance with Section 1.08, "Acceptance of Work" in these General Specifications as follows:

The materials incorporated into the riprap and slope protection shall be sampled, tested and evaluated in accordance with the specifications and test methods referenced in Subsection 6.06.2, "Materials" in these General Specifications. The installation of the riprap and slope protection will be accepted in accordance with Subsection 1.08.4, "Measured or Tested Conformance" in these General Specifications.

6.06.5 METHOD OF MEASUREMENT. Loose Stone Riprap, Grouted Stone Riprap, Sacked Concrete Revetment, Gabions, and Wire Enclosed Riprap Slope Protection shall be measured by the cubic meter based on the dimensions shown on the plans or ordered in writing by the Engineer, for all quantities authorized, constructed and accepted by the Engineer.

Reinforced Concrete Slope Protection, Wire Mesh Slope Protection and Paving Tile Slope Protection, Geotextile Fabric (if included in the Bill of Quantities) will be measured by the square meter based on the dimensions shown on the plans or ordered in writing by the Engineer, for all quantities authorized, constructed, and accepted by the Engineer. When Geotextile Fabric is required by the plans or specifications, but not included in the Bill of Quantities, it shall be considered subsidiary. When it is included in the Bill of Quantities it shall be as specified, measured, and paid for under Section 6.16, "Geotextiles" in these General Specifications. No separate measurement will be made for the anchorage system for Wire Mesh Slope Protection which shall be considered subsidiary to that Item of Work.

Excavation below original ground elevations for riprap and other forms of slope protection shall be measured in accordance with Paragraph 2.03.2.3 "Roadway Excavation - Channel and Ditch" in these General Specifications.

6.06.6 PAYMENT. The amount of completed and accepted Work as measured above will be paid for at the contract unit price(s) specified in the Bill of Quantities which price(s) shall be full compensation for furnishing all materials, for all labor, equipment, tools, supplies, and all other items necessary for the proper completion of the Work as specified in Subsection 1.07.2, "Scope of Payment" in these General Specifications.

PAYMENT WILL BE MADE UNDER ONE OR MORE OF THE FOLLOWING:

ITEM NO	PAY ITEM	PAY UNIT
60601	Loose Stone Riprap	Cubic Meter
6060101	Loose Stone Riprap, Class A	Cubic Meter
6060102	Loose Stone Riprap, Class B	Cubic Meter
6060103	Loose Stone Riprap, Class C	Cubic Meter
6060104	Loose Stone Riprap, Class D	Cubic Meter
6060105	Loose Stone Riprap, Class ___	Cubic Meter
60602	Grouted Stone Riprap	Cubic Meter
6060201	Grouted Stone Riprap, Class A	Cubic Meter
6060202	Grouted Stone Riprap, Class B	Cubic Meter
6060203	Grouted Stone Riprap, Class C	Cubic Meter
6060204	Grouted Stone Riprap, Class D	Cubic Meter
6060205	Grouted Stone Riprap, Class ___	Cubic Meter
60603	Sacked Concrete Revetment	Cubic Meter
60604	Gabions	Cubic Meter
60605	R.C. Slope Protection	Square Meter

60606	**Wire Mesh Slope Protection**	**Square Meter**
60607	**Paving Tile Slope Protection**	**Square Meter**
60608	**Wire Enclosed Riprap Slope Protection**	**Cubic Meter**